



Work Breakdown Structure (WBS)



1. A fundamental project management method which uses a hierarchical tree structure.
2. It defines and organize the overall project scope
3. Each level of WBS has to cover 100% of the project(classified based on effort not duration)
4. Main focus is Outcome

1.100% Rule:

It means In each Level 100% of product outcome gaining

2.Planned Outcomes,Not Planned Actions:

Action can be made to any project more than 100% or less than 100% of the project depending on the number of actions needed.To prevent this problem,outcome based breakdown makes a project independently broken and more successful.

3. Level 2 is the Most Important:

It affects how real prices and schedule data are organized for next project cost and schedule estimation.

4. The Four Elements in Each WBS Element:

1. The scope of work
2. Time frame
3. Budget frame
4. Responsible HR

5. Mutually-exclusive Elements:

A WBS must not have any scope definition overlap between its various components thus every single component is independent and unique.

6. How Far Down?

When it is no longer possible to define planned outcomes

6.1 The 40-Hour Rule of Decomposition:

Typically, there is no need to further break down a project once each element has around 40 hours(work week) of direct labor committed.

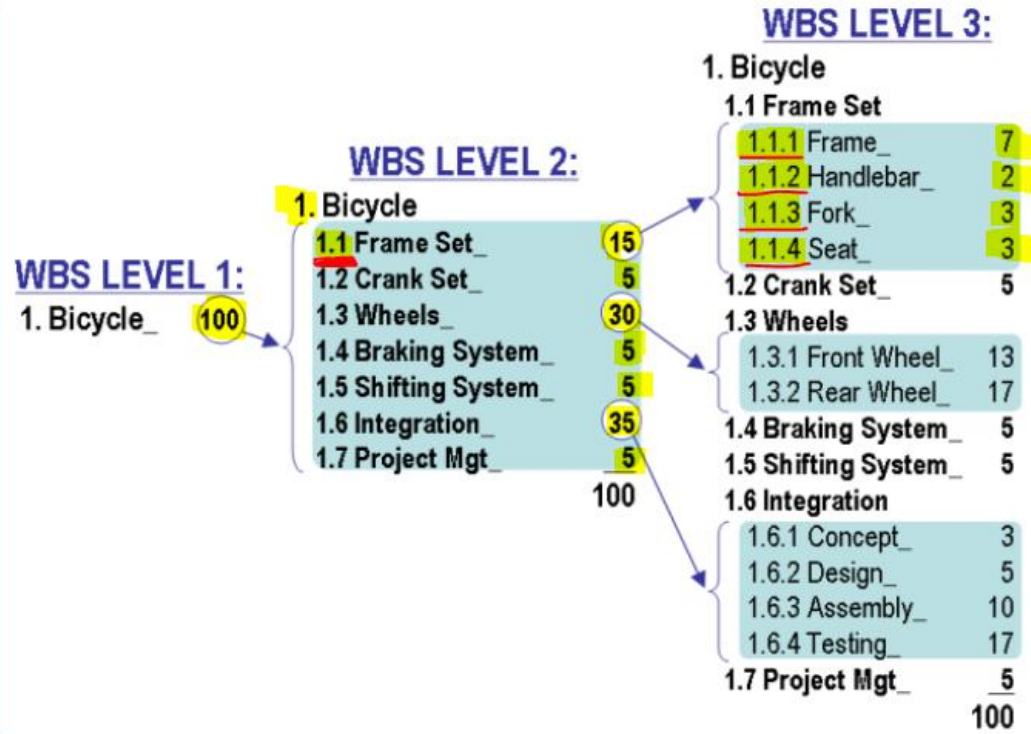
6.2 The 4% Rule of Decomposition:

This guideline states that a WBS is sufficiently decomposed when the lowest element represents around 4% of the whole project.

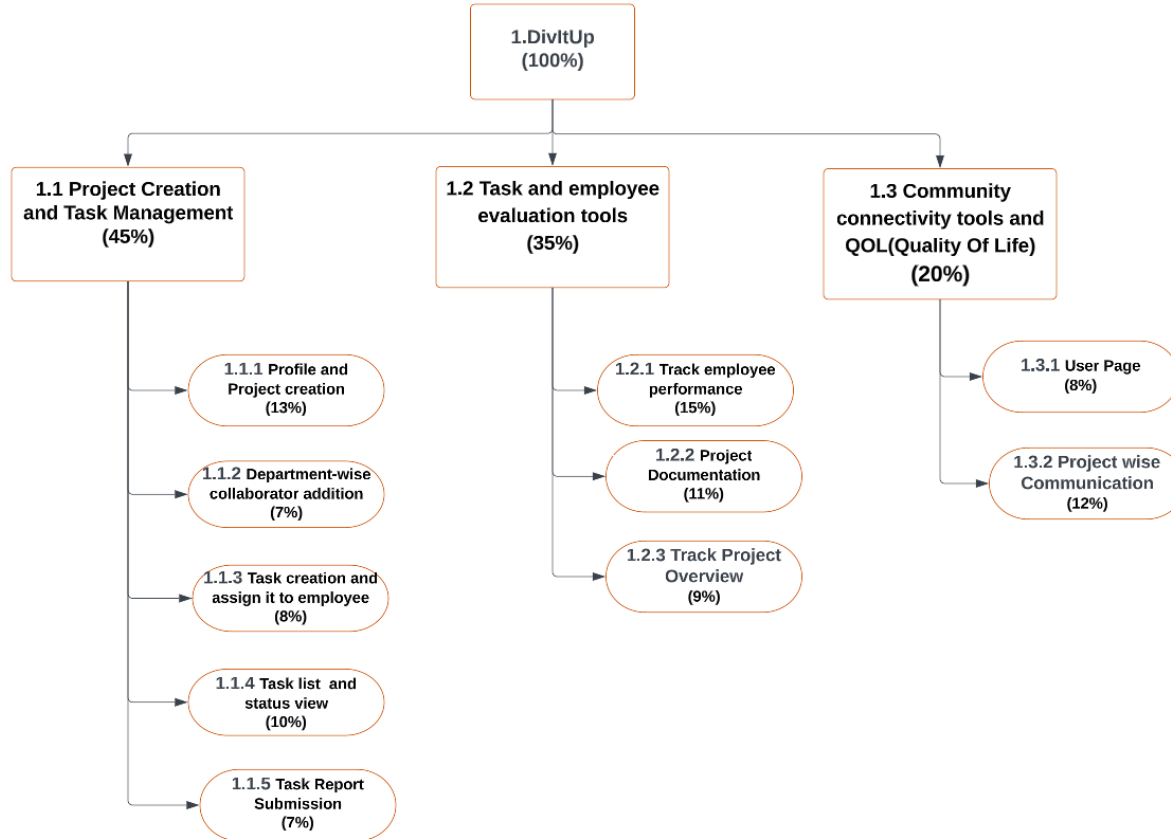
7.WBS Identification Numbering:

To show the hierarchical structure, WBS elements usually have an identification number. Since there are three digits separated by a decimal point, for instance, 1.3.2 Rear Wheel designates this item as a Level 3 WBS component.

Example: 100% rule and naming Identification Numbering



WBS Of DivItUp



I.I.3 TASK CREATION AND ASSIGN IT TO EMPLOYEE

8%

1. Deliverables:

- Create tasks
 - Task info
 - Deadline
- Assign task to member of project
- Assign comments to tasks

2. The beginning and end dates for the scope of work.

Start date: 1-03-23

End date: 8-03-23

3. The budget for the scope of work.

Budget = hours of labor x average hourly salary of developers + overhead
= $48 \times 320 + (48 \times 320) = 30720$

4. The name of the person responsible for the scope of work: X

I.I.4 TASK LIST AND STATUS VIEW

10%

1. The scope of work, including any “deliverables.”

- View project specific tasks
- Scrum board

2. The beginning and end dates for the scope of work.

Start date: 8-03-23

End date: 17-03-23

3. The budget for the scope of work.

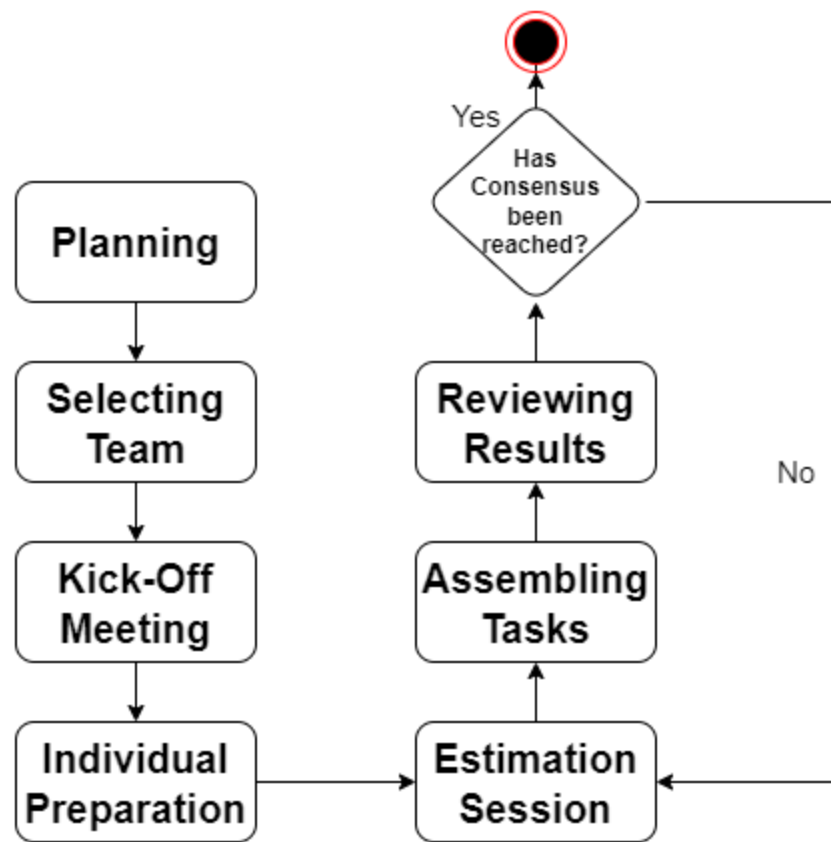
Budget = hours of labor x average hourly salary of developers + overhead
= $60 \times 320 + (60 \times 320) = 38400$

4. The name of the person responsible for the scope of work: Z



Delphi Wideband Method





Wideband Delphi Technique is a consensus-based estimation technique for estimating effort. Useful when estimating time to do a task. Participation of experienced people and individually estimating would lead to reliable results.

- ❑ **Choosing the team**
- ❑ **Kick-off Meeting**
- ❑ **Individual Preparation**
- ❑ **Estimation Session**
- ❑ **Assemble tasks**
- ❑ **Review result**

Details of Different Steps

- 1. *Choosing the team***- The project manager selects the estimation team and a moderator. The team should consist of 3 – 7 members. The team should include representatives from every engineering group that will be involved in the development of the work product being estimated.
- 2. *Kickoff meeting***- The moderator prepares the team and leads a discussion to brainstorm assumptions, generate a WBS, and decide on the units of estimation.
- 3. *Individual preparation***- After the kickoff meeting, each team member individually generates the initial estimates for each task in the WBS, documenting any changes to the WBS and missing assumptions.

Details of Different Steps

4. Estimation session- The moderator leads the team through a series of iterative steps to gain consensus on the estimates. At the start of the iteration, the moderator charts the estimates on the whiteboard so the estimators can see the range of estimates. The team resolves issues and revises estimates without revealing specific numbers. The cycle repeats until either no estimator wants to change his or her estimate, the estimators agree that the range is acceptable, or two hours have elapsed.

5. Assembling tasks- The project manager works with the team to collect the estimates from the team members at the end of the meeting and compiles the final task list, estimates, and assumptions.

6. Reviewing results- The project manager reviews the final task list with the estimation team.

INDIVIDUAL PREPARATIONS

Task list

Tasks to achive goal	Time
_____	_____
_____	_____
_____	_____
_____	_____
_____	_____
_____	_____
Calendar waiting time, delays	_____
_____	_____
_____	_____
_____	_____
_____	_____
_____	_____
Project overhead tasks	_____
_____	_____
_____	_____
_____	_____

Assumptions

1. _____

2. _____

3. _____

4. _____

5. _____

6. _____

7. _____

8. _____

9. _____

10. _____

11. _____

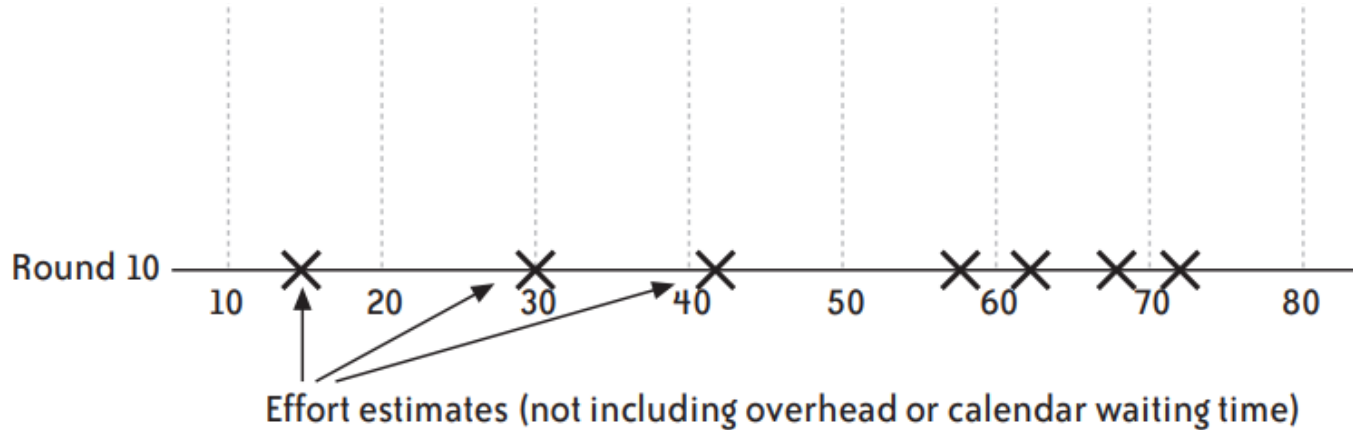
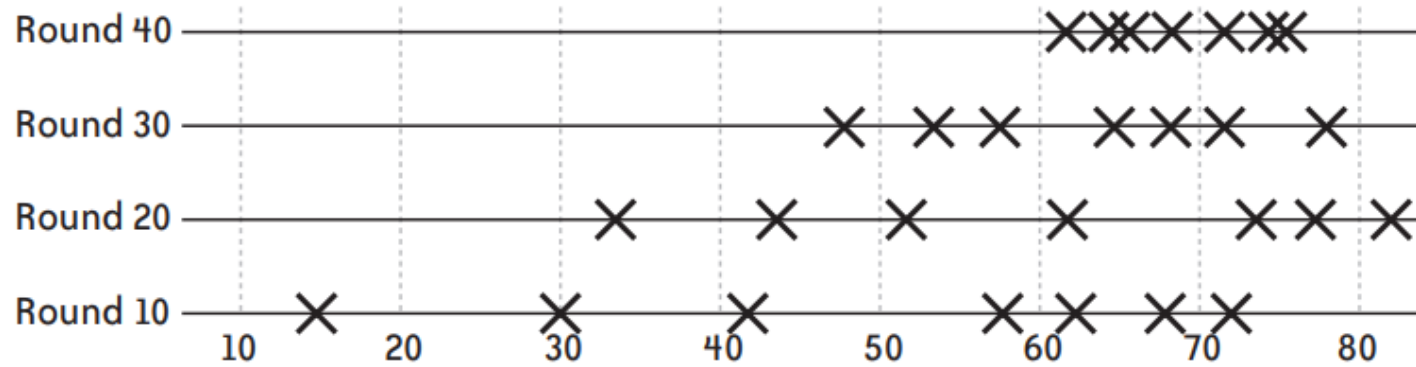
12. _____

13. _____

14. _____

15. _____

ESTIMATION SESSION



ASSEMBLE TASKS

Goal statement <u>To estimate the time to develop prototype for customers A & B</u>									
Estimators <u>Mike, Quentin, Jill, Sophie</u>								Units <u>days</u>	
<i>Shaded items must be discussed</i>									
WBS# or priority	Task name	M.	Q.	J.	S.	Best-case	Worst-case	Avg.-hi & lo	Notes
1	Interview customers (A+B)	6	4	3	3	3	6	3.5	
2	Develop requirements docs	5	10	2	5	2	10	5	Discrepancy between Q. and J.
3	Inspect requirements docs	7	5	6	5	5	7	5.5	
4	Do rework	8	7	9	7	7	9	7.5	
5	Prototype design	28	23	31	25	23	31	26.5	
6	Test design	9	7	6	6	6	9	6.5	
	Total	63	56	57	51	46	72	54.5	

Pros and Cons

Advantages

- ❑ Wideband Delphi Technique is a consensus-based estimation technique for estimating effort.
- ❑ Participation of experienced people and they individually estimating would lead to reliable results.
- ❑ Anonymity maintained throughout makes it possible for everyone to express their results confidently.
- ❑ Assumptions are documented, discussed and agreed.

Disadvantages

- ❑ Management support is required.
- ❑ The estimation results may not be what the management wants to hear.